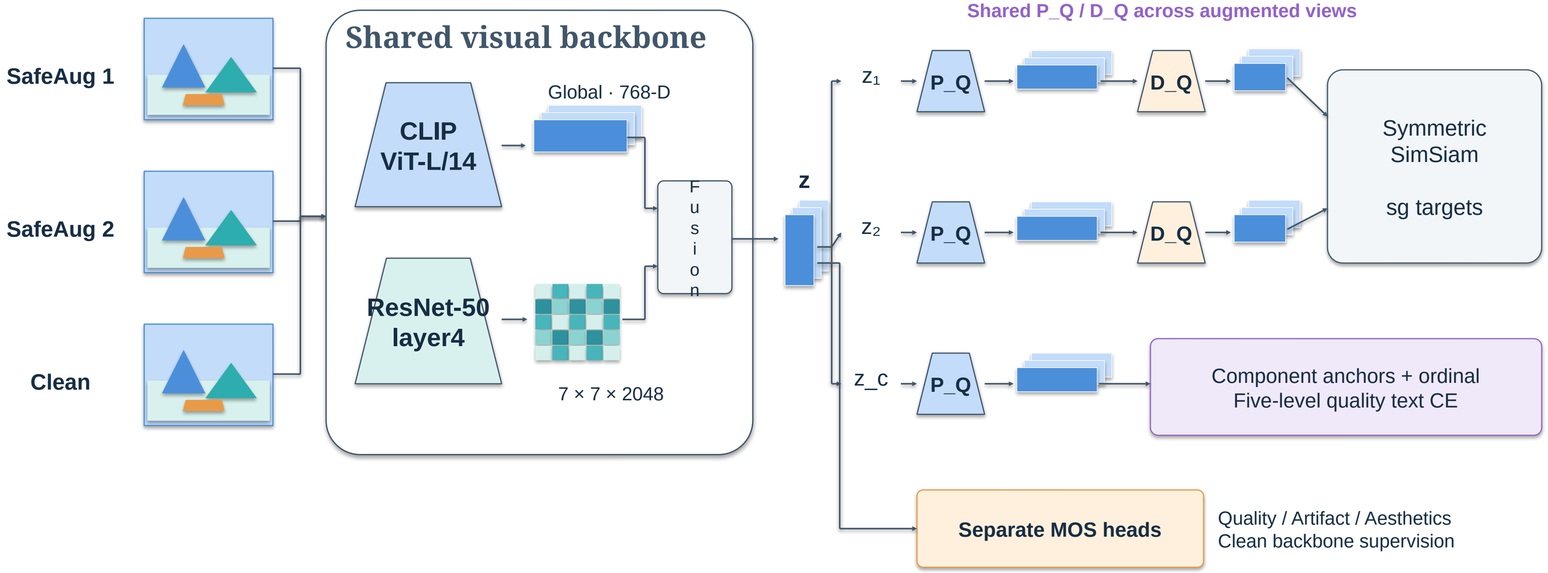
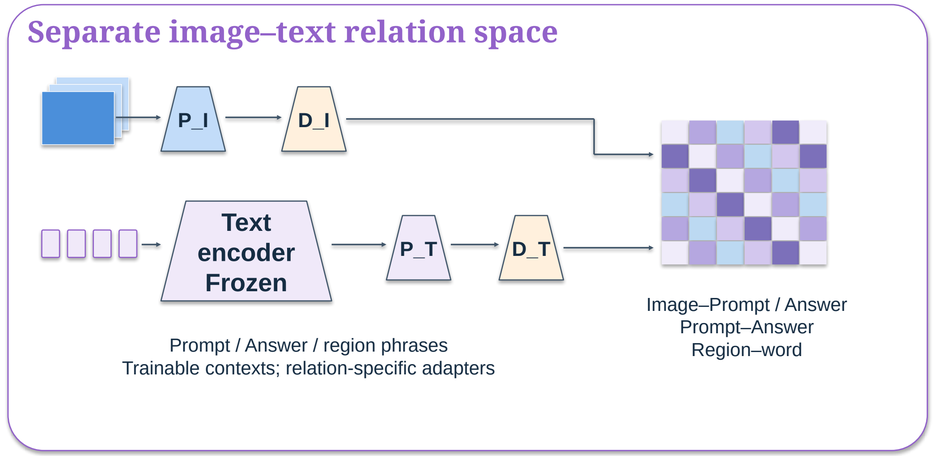
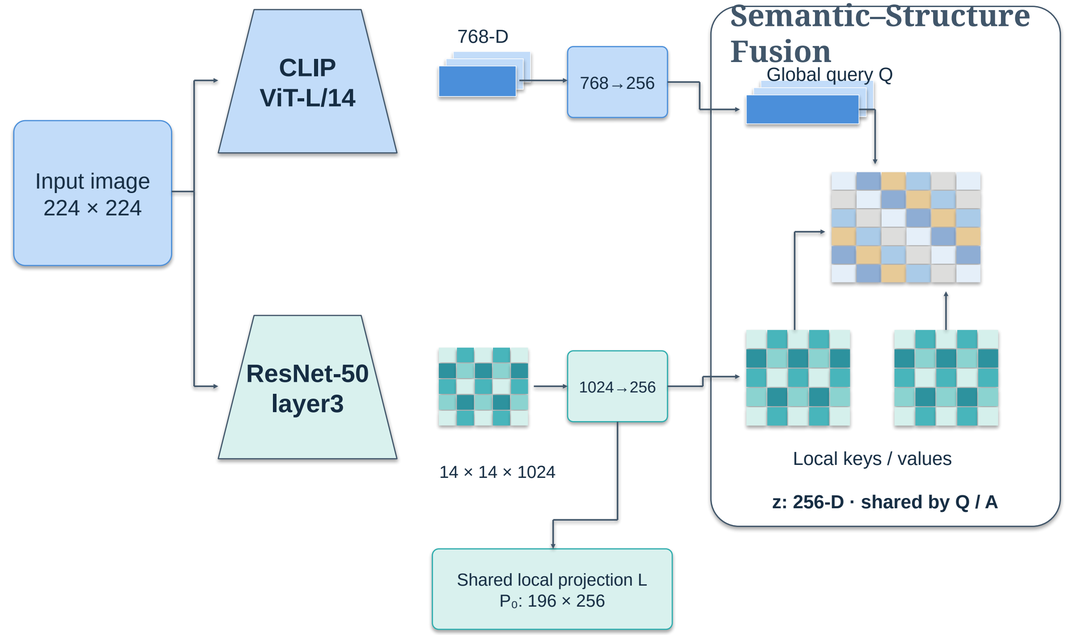
(a) Quality-aware multimodal pretraining

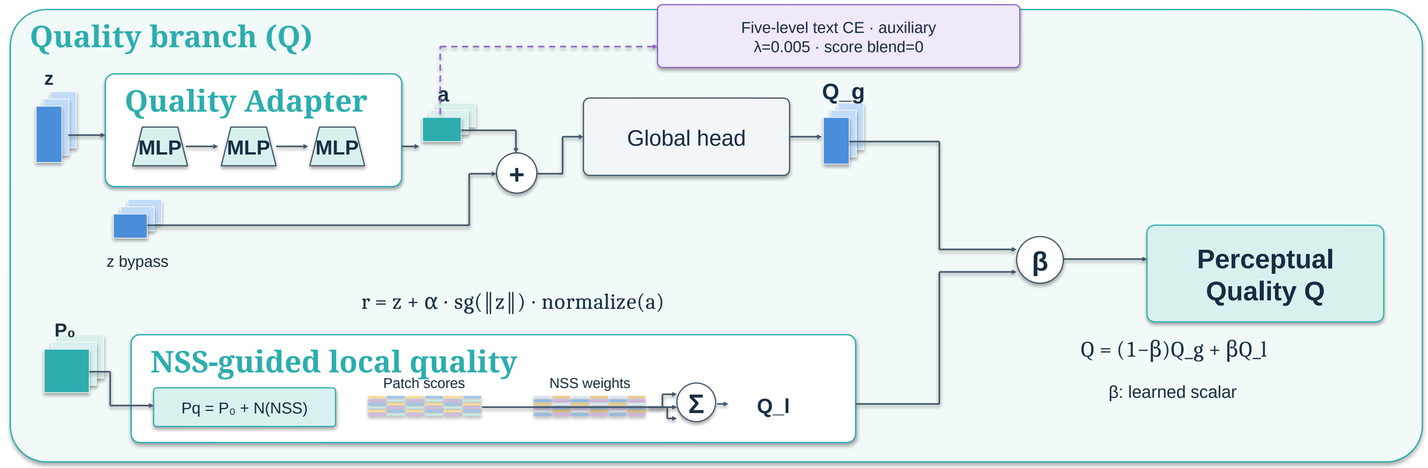




P_Q → Quality Adapter initialization
Backbone / contexts / alignment adapters / quality bank transferred

(b) Downstream quality & alignment assessment





Same z / P₀ / u ports denote the same feature tensors. sg: stop-gradient. Visual encoders trainable; text backbone frozen. Source layer4 → downstream layer3: local projection reset.

(a) Quality-aware multimodal pretraining

SafeAug 1
SafeAug 2
Clean

Shared dual visual encoder
CLIP ViT-L/14
global feature (1 × 768)
ResNet-50 (layer4)
local feature (7 × 7 × 2048)
Fusion
z

shared P_Q weights
P_Q → q_1 → D_Q → sg(q_2)
SimSiam (symmetric)
P_Q → q_2 → D_Q → sg(q_1)
P_Q → q_c

Component anchors + ordinal
5-level CE (1 2 3 4 5)

MOS heads (from backbone features)
MOS supervision (1 2 3 4 5)

Image tensor 224 × 224 × 3
P_I → D_I
"a red boat on a lake"
trainable context tokens
Text encoder (frozen)
P_T → D_T
Image–text relations

P_Q → Quality Adapter (parameter transfer)

Backbone / contexts / alignment adapters transferred; scoring heads reset
- - - → : parameter transfer
- - - → : auxiliary loss (training only)

(b) Downstream quality & alignment assessment

Input image 224 × 224 × 3
CLIP ViT-L/14
global feature (1 × 768)
768 → 256
g
ResNet-50 (layer3)
local feature (14 × 14 × 1024)
1024 → 256
P_0 (196 × 256)
layer3 projection newly initialized

Semantic–Structure Fusion
global query (g) → Q
Attention
local key/value (P_0) → K, V
z (256)

Quality branch (Q)
Quality Adapter
z256 → MLP → MLP → MLP → a
a → bottom five-level CE (auxiliary only)
5-level text CE + training only blend=0 (1 2 3 4 5)
Global head
MLP → Q_g
$r = z + \alpha\ \mathrm{sg}(\|z\|)\ \mathrm{normalize}(a)$

P_0 (196 × 256)
NSS-guided local quality
NSS (image)
Patch scores
NSS weights
→ Q_l

$Q = (1-\beta)Q_g + \beta Q_l$
β : learned scalar
Perceptual Quality Q

Alignment branch (A)
z256 → Image adapter A_G → v
Global pair head
u u → MLP → A_g
sample-wise γ (conditioned on $\mathbf{v}$, $\mathbf{u}$)
$A = \gamma A_g + (1-\gamma) A_l$
Text–Image Alignment A

"a red boat on a lake"
context tokens
Text encoder (frozen)
u
Text adapter A_T → u
Local–text matching
P_0 (196 × 256)
similarity
attention pooling
Local pair head
MLP → A_l

Repeated z / P_0 symbols denote shared features.

编辑说明 / Editing guide

第 1 页：可编辑重绘版。文字、公式、模块、箭头、特征张量和热图均为原生 PPT 对象。

第 2 页：原始论文风格图，仅作视觉参考。 PNG 本身不能在 PowerPoint 中拆成原生图层。

在第 1 页直接点击文字修改；选中连接线后可拖动端点。图形填充、线条和字体可直接修改。

张量与热图以局部组合保存：右键 → 组合 → 取消组合，可编辑每个小格；也可 Ctrl+ 单击选择组内对象。

全选多个对象后可组合移动。重复的 z / P_0 / u 是同一个特征的命名端口，用于避免长线交叉。

已按代码修正关键关系： $z_\rightarrow$ 全局 Adapter ， $P_{0\rightarrow}$ 局部匹配；质量标量 β ，一致性逐样本 γ 。

保留原图配色和双面板视觉语言，但原生重绘并非与原 PNG 逐像素一致。

导出论文图：文件 → 导出 → PDF ；或另存图片。排版时建议保留矢量 PDF 。

依据：代码 e127834 ； CLIP Large 、下游 layer3 、 Residual Quality Adapter 、 CE=0.005 、标量质量融合。

未纳入实验中的逐图质量门控，也未把学习率消融宣称为已确定主方法。